

Journal Paper Summary (25 points total)

Paper Title	User-Centered Design and Interactive Health Technologies for Patients
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1. What do you think the paper is about in layman's terms? List all the benefits of using user-centered design (UCD) that are mentioned in the paper. (UCD) [5 points]

Focus of this question
• This question encourages you to evaluate the arguments, evidence, assumptions, and conclusions about key issues (i.e. think critically about the paper) • This question encourages you to develop your own knowledge, comprehension and conceptual understanding and to connect, synthesize, and/or transform your ideas into a new form (i.e. be a creative thinker and contribute your ideas and thoughts) • This question challenges you to consider diverse perspectives (gender, political, ethnic, racial, etc) during class or in assignments

Write your answer below:

This paper looks at involving patients in the design and development process of an application customized to patient's needs. To do this, the study developed a "Pocket Personal Assistant for Tracking Health (Pocket PATH)" to involve patients with the design, testing and increasing the likelihood of promoting the intended health outcomes.

Some of the benefits of involving patients with the design of applications are:

- Ensuring functionality and usability increasing the likelihood of promoting the intended health outcomes.
- Reduce development time because usability problems are identified and resolved before the systems were launched.
- Implicit benefits show up as justification for design choices.

2. What are the three guiding principles of user-centered design (UCD) introduced by Gould and Lewis? In your own words, summarize how each of these principles of user-centered design were applied to the design and testing of Pocket PATH [5 points]

Focus of this question
• This question encourages you to connect your learning to “real world” issues or life experiences and consider diverse perspectives for the application of concepts in the paper to the real-world • This question encourages you to develop your own knowledge, comprehension and conceptual understanding and to connect, synthesize, and/or transform your ideas into a new form (i.e. be a creative thinker and contribute your ideas and thoughts)

Write your answer below:

The three guiding principles of UCD from Gould and Lewis are:

1. Early Focus on Users and Tasks: Understand user base and their interests.
2. Empirical Measurement: Factors that contribute to overall usability and should be measured and analyzed.
3. Iterative Design: A cycle of design, test, measure and redesign that should be repeated.

The first point “early focus on users and tasks” is pretty evident in this paper. In the introduction, Dabbs et al. established that they would be working with patients to develop Pocket PATH throughout the entire design process. To do this, they “gathered information about background characteristics for the patients that could affect their use of computer-based technologies from a variety of sources.” They then recruited these patients and made sure that the patients they selected were representative of the larger collective to ensure that the app could benefit patients on any spectrum (this spectrum being patients with recent lung transplants, patients less than 18 years old, English speaking, etc.). The study established what tasks were essential for this application such as monitoring complications, measuring lung volumes, etc. and established a way to monitor the patient’s self-performed maintenance tasks. Finally, the study established a medium to keep track of patient’s self-monitoring with a hand-held device.

In the second point, “empirical measurement”, the study identified factors that were used to measure task completion, identify user frustrations and usability issues by surveying users to further investigate the source of frustrations. To accomplish this, three UCD sessions were conducted in a controlled environment to obtain feedback from the users. The study

team stuck with a very iterative design process (which I will cover later) in which they repeated tiny steps of designing the application, constantly receiving user feedback until requirements were met. Some of these steps were getting feedback on initial prototype design and interface sketches, having users simulate tasks and measuring performance on these tasks, and finally, using taped sessions to reveal potential errors and problems for future modifications. Combining all these methods, the study team effectively empirically measured the patient's performance, listened to feedback, etc. to make future design decisions.

In the third and final point which was "iterative design", which was intertwined with the second point "empirical measurement", the study team took patient feedback to make small improvements to the Pocket PATH app with user feedback in mind. The study team accomplished this by taking Pocket PATH out of a controlled environment and having the patients test this in their homes to assess functionality of the features, data recording, etc. Throughout this process, the patients were given a technician on-call number, extensive instructions on how to use the hand-held and tasked to record daily measurements and vitals. Every day, the patients would fill out the data charts, and if an anomaly occurred, the technical team was prompt to respond to them. At the end of the session, six test users were prompted to test the final version and asked to provide feedback to evaluate the features. They were also given a Post-Study System Usability Questionnaire to assess user satisfaction. This is the final point of the three guiding principles which the study implemented perfectly.

The authors offer possible explanations for why developers of interactive health technologies (IHTs) fail to incorporate user-centered design (UCD) or fail to report how the principles of UCD were integrated into the development of their IHTs. Think about, list and discuss five (5) other possible explanations NOT mentioned in the paper why developers fail to incorporate or report the inclusion of UCD in the build out of IHTs [15 points]

Focus of this question
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Write your answer below:

The authors list the following reasons as to why developers fail to incorporate user-centered design:

- Lack of appreciation for the importance of usability testing
- Lack of time and resources to devote to upfront research and development
- Limited expertise in the principles and techniques of UCD
- Tendency to develop IHTs based on developer driven needs and priorities rather than the users’ needs and priorities.

I think some other reasons developers don’t want to incorporate UCD into IHTs are:

- Measurement of success is not how much a user is liking the app, but rather the collectible data at the end of the day.
- Deadlines are also another reason that may deter researchers and software developers from integrating UCD practices into designing IHTs because and iterative approach to design is a time consuming one.
- Legal concerns. Maybe the data collected and the way of collection violate certain laws (HIPPA, etc.) that should be kept confidential to each patient.
 - o Counterpoint: I had to work with this during my summer job, and there are ways to still have good UCD practices while not violating HIPPA laws.
- Lack of funding for creating a team of software developers, scientists, etc. This study had the resources to dedicate an entire team to UCD practices, but not every team will have this luxury.

- The patient population of this study were competent enough to engage with such tasks, but other sample sizes of patients and users might not always be the most reliable user base.